Thin-z TDE Outer-Boundary Comparison
Extrapolated Sommerfeld versus Zero-Rate Characteristic CPBC

AthenaK Z4c validation

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Abstract

We compare two no-sponge tidal-disruption-event evolutions with identical initial data, mesh, executable, MPI decomposition and placement policy, and fourth-order boundary extrapolation. The only intentional physical or numerical boundary difference is the outer-boundary right-hand side: extrapolated Sommerfeld or the archived zero-rate characteristic constraint-preserving boundary condition (CPBC). Sommerfeld develops a lower-z-localized instability at $t/M \simeq 7$ and becomes nonfinite at $t/M = 7.2$. The CPBC evolution remains finite through $t/M = 30$, with decreasing global constraint norms and a bounded close-face Hamiltonian maximum. Its orbital-plane density slices retain a coherent, nearly circular stellar core throughout the run. Thus the archived zero-rate CPBC cures the observed instability in this configuration. This result does not validate the newer experimental tangential-principal implementation, and a matched far-boundary reference is still required for formal reflection, density, and trajectory acceptance gates.

1 Question and answer

For the tested thin-z TDE configuration, the answer is yes: the archived zero-rate CPBC prevents the instability exhibited by extrapolated Sommerfeld. The conclusion is empirical and configuration-specific. It is supported by a byte-identical executable comparison and by survival for more than three nominal fastest-gauge round trips.

2 Matched setup

Both evolutions used executable SHA-256

`cd7cefef042ff075e85688dd3ec08dca243df479d50b91e0439dd8a20da46478`.

The domain and numerical settings were:

Quantity	Value
Domain	$x \in [-48, 72]M$, $y \in [-48, 48]M$, $z \in [-6, 42]M$
Initial star center	approximately $(40, 0, 0)M$
Root spacing	$0.75M$
Target time	$30M$
Close physical face	lower z
Boundary extrapolation	order 4
Outer sponge	explicitly disabled
Resources	32 Aurora nodes, 12 MPI ranks per node

The Sommerfeld reproducer was job `8718377`. The CPBC evolution used fresh segment `8718576` and restart segments `8718778`, `8718933`, and `8719210`. Every CPBC segment exited with status zero. Each restart generation contained exactly 384 nonempty rank-local members.

3 Evolution outcome

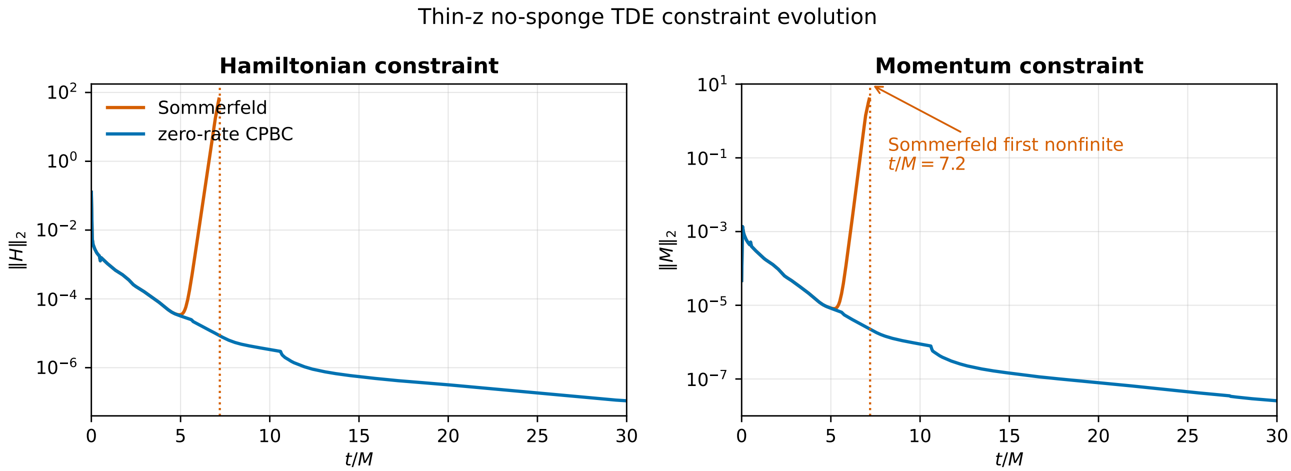

Figure 1: Global Hamiltonian and momentum constraint histories. The two curves agree before the causal return. Sommerfeld then grows catastrophically and becomes nonfinite at $t/M = 7.2$, while CPBC continues to $t/M = 30$ with decreasing norms.

The first material Sommerfeld growth appears at the lower-z face:

t/M	$\max\|H\|$	$(x, y, z)/M$
7.00313	31.8178	$(-0.0703, 0.4453, -5.9766)$
7.05234	39.3038	$(-0.0703, 0.4453, -5.9766)$
7.10156	44.4944	$(2.1797, -0.0703, -5.9766)$
7.15078	69.0354	$(2.0391, 0.0703, -5.9766)$

The first nonfinite history row is at $t/M = 7.2$. Shortly afterward the stellar density tracker falls to the atmosphere floor.

At the matched finite history sample $t/M = 7.1015625$, the global comparison is:

Diagnostic	Sommerfeld	CPBC	Sommerfeld/CPBC
$\|H\|_2$	4.76992×10^{1}	9.00521×10^{-6}	5.297×10^{6}
$\|M\|_2$	3.13449	2.37331×10^{-6}	1.321×10^{6}
$\|\Theta\|$	4.32375×10^{-5}	1.30143×10^{-8}	3.322×10^{3}
$\max\|\Theta\|$	3.85857×10^{-2}	1.80522×10^{-5}	2.137×10^{3}
$\max\|\delta\alpha\|$	1.08070×10^{-2}	9.48888×10^{-6}	1.139×10^{3}
$\max\|\delta\beta^i\|$	3.24569×10^{-4}	3.58307×10^{-6}	9.058×10^{1}
$\max\|\delta\widetilde{\Gamma}^i\|$	3.20782×10^{-3}	2.15356×10^{-5}	1.490×10^{2}

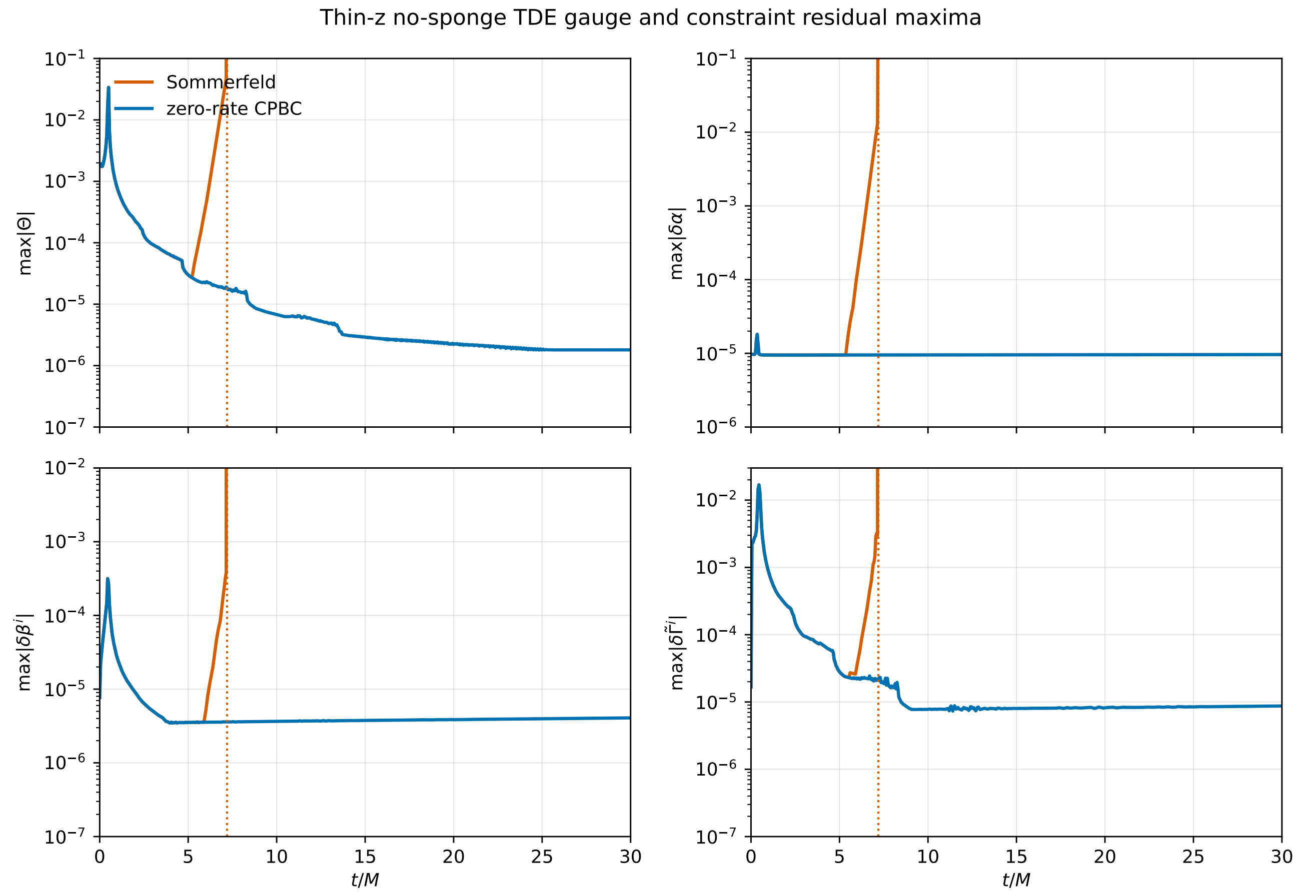

Figure 2: Gauge and constraint residual maxima. Fixed vertical ranges retain the pre-failure CPBC baseline and the Sommerfeld onset; catastrophic Sommerfeld values beyond the displayed range clip at the upper axis.

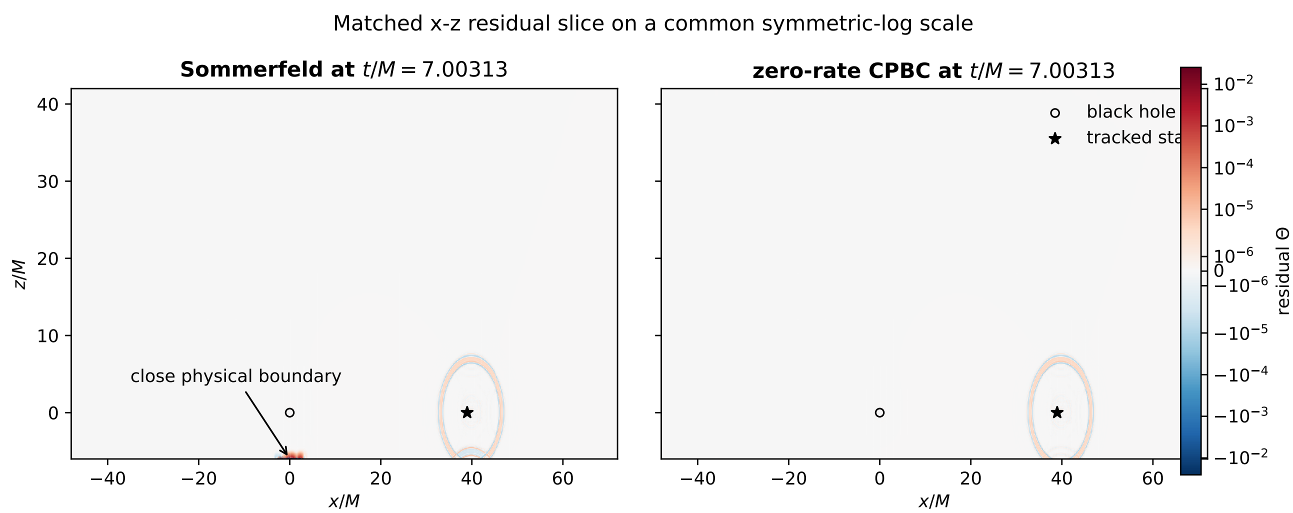

Figure 3: Matched x-z residual Θ slices at $t/M = 7.00313$ on a common symmetric-logarithmic scale. Sommerfeld has already developed a strong localized feature at the close lower-z boundary near the black hole. The CPBC slice remains bounded.

3.1 Orbital-plane stellar morphology

The available fluid slices resolve the x–y orbital plane. We measured all 61 CPBC density outputs rather than inferring shape from $\rho_{\max}$ alone. At each time, the dense-core region is defined by $\rho \geq \rho_{\max}/2$; its area gives $R_{1/2} = \sqrt{A_{1/2}/\pi}$, and a density-weighted covariance gives the principal-axis ratio a/b. A peak-centered, peak-normalized density map provides a separate shape-L_2 diagnostic.

Diagnostic	Initial	Range through $30M$	Final
$\rho_{\max}$	1.22674×10^{-4}	-0.10% to $+4.89\%$	1.28672×10^{-4}
$R_{1/2}/M$	0.08363	-0.63% to $+1.24\%$	0.08311
a/b	1.0065	1.0005 to 1.0748	1.0282
Centered shape L_2	0	0 to 0.0738	0.0442

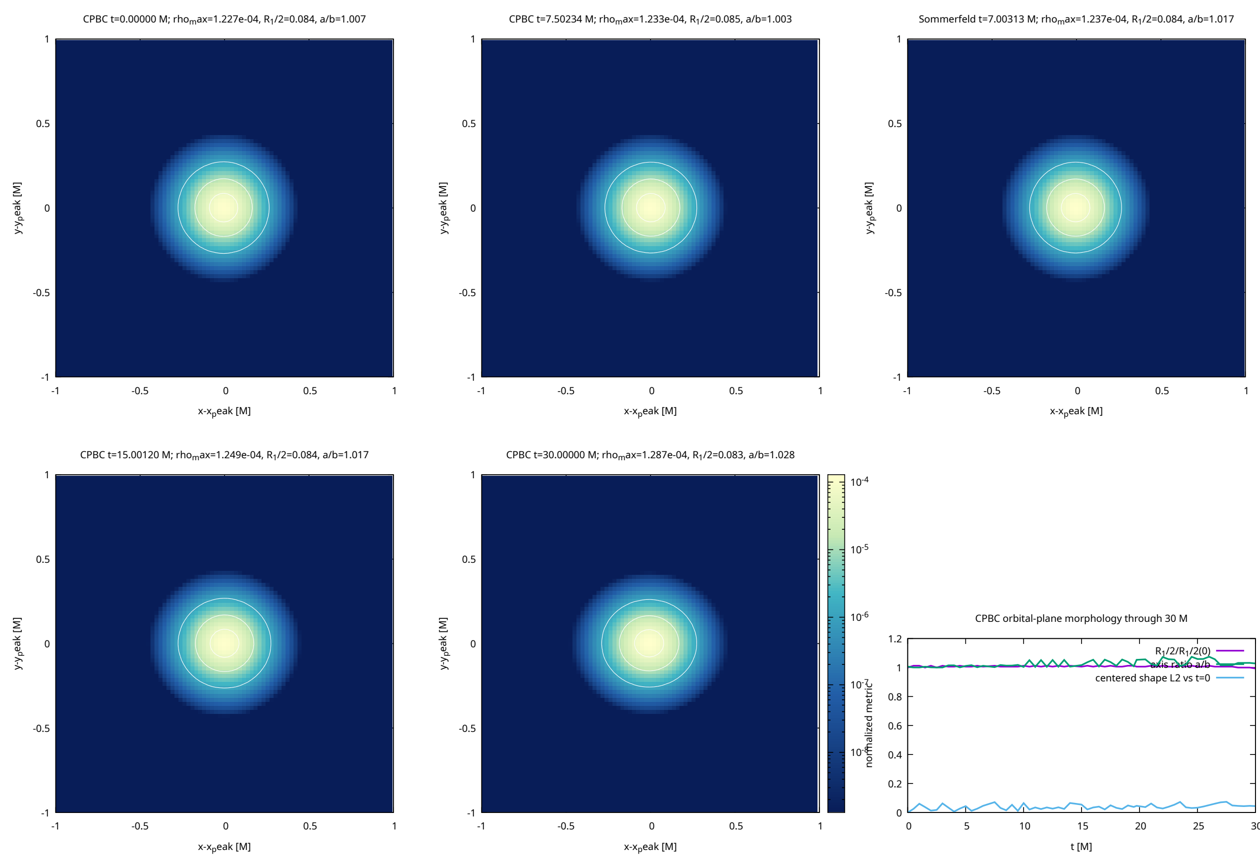

Figure 4: Peak-centered orbital-plane density with a common absolute logarithmic color scale. White contours mark 50, 10, and 1 percent of each panel's peak. The lower-right panel summarizes all 61 CPBC outputs. Sommerfeld is shown at $t/M = 7.00313$, when the boundary failure is already material but the star remains physical. No CPBC density slice was stored at that exact time; the nearest displayed CPBC slice is $t/M = 7.50234$, offset by $0.49921M$.

The Sommerfeld slice was deliberately chosen before its density becomes floor-dominated: it has $\rho_{\max} = 1.23677 \times 10^{-4}$ and close-face $\max |H| = 31.8178$. Its $R_{1/2} = 0.08415M$ and $a/b = 1.0169$ remain stellar. The nearest CPBC slice has $R_{1/2} = 0.08467M$ and $a/b = 1.0029$; their centered normalized density maps differ by 0.0677, with the stated $0.49921M$ time offset.

These measurements support a positive, qualified answer to the shape question: the CPBC run preserves a single coherent and resolved orbital-plane stellar core through $30M$, with at most 7.5 percent ellipticity in this diagnostic. They do not establish the unobserved vertical profile or full three-dimensional shape, nor do they replace the still-open far-reference density and trajectory gates.

4 CPBC state at the target time

The CPBC case reaches $t/M = 30$ with no nonfinite, bad-metric, MPI, SYCL, GPU, or restart error. Its final diagnostics are:

Diagnostic	Value at $t/M = 30$
$\|H\|_2$	1.09007×10^{-7}
$\|M\|_2$	2.56218×10^{-8}
Close-face max $\|H\|$	1.20496×10^{-4}
$\rho_{\max}$	1.28672×10^{-4}
Tracked star position	$(35.3731, -0.00586, -0.00586)M$
Bad-metric count	0

An independent late-time projection over 13 slices in $t/M \in [27, 30]$ reports all ten incoming characteristic families:

Family	Maximum RMS	Family	Maximum RMS
Lapse	3.22520×10^{-6}	Longitudinal shift	3.45682×10^{-6}
Transverse shift 1	1.83089×10^{-6}	Transverse shift 2	8.37305×10^{-9}
Scalar constraint Θ	6.41182×10^{-6}	Scalar constraint Z	1.07866×10^{-6}
Transverse constraint 1	1.69201×10^{-6}	Transverse constraint 2	1.08745×10^{-8}
TT plus	1.34214×10^{-6}	TT cross	8.72326×10^{-8}

All remain finite and $O(10^{-6})$ or smaller over $27 \le t/M \le 30$. Incoming and outgoing rows are reconstructed from the full evolved conformal metric and independently transcribed characteristic formulae; boundary-enforcement diagnostics are not used as reflection measurements.

5 Analysis safeguards

Restarted outputs duplicate the state at each segment boundary. The reader finds duplicate slice times at $6.50039M$, $14.0027M$, and $19.7508M$. Each pair has identical mesh layout and exactly zero field difference, although binary header metadata differs. The analysis collapses only field-identical duplicates and rejects conflicting duplicates.

The lower-face Sommerfeld metric becomes sufficiently distorted that the face-normal frame develops an out-of-x-z-plane component of 1.00845×10^{-3}. Because an x-z slice cannot supply the missing y derivative, the post-failure all-ten face projection is rejected rather than evaluated with a coordinate-aligned approximation. Raw coordinate-bearing extrema still localize the instability at the first lower-z cell layer.

6 Scope and limitations

This report validates the archived zero-rate executable in the selected TDE comparison. The source branch also contains a newer experimental tangential-principal target. That newer implementation was not used in these TDE jobs and must not inherit this validation claim.

The existing far-boundary reference has different thin-z geometry, different residual-slice coverage, and no executable parity through $t/M = 30$. Therefore this comparison does not yet measure a far-reference-subtracted returning characteristic signal. The formal tenfold gauge/constraint reflection gate, the per-family 10 percent non-regression gate, the one percent density gate, and the $0.1R_\star$ trajectory gate remain open. A matched far-boundary run with x-z residual output is required for those claims.

7 Reproducibility

The figures are regenerated by:

```
python3 analysis/z4c_characteristic/plot_tde_boundary_comparison.py \
  --sommerfeld <sommerfeld-run> \
  --cpbc <cpbc-run> \
  --output-dir docs/z4c/figures --formats pdf
python3 analysis/z4c_characteristic/plot_tde_stellar_morphology.py \
  --sommerfeld-run <sommerfeld-run> \
  --cpbc-run <cpbc-run> \
  --output docs/z4c/figures/tde_stellar_morphology_cpbc_vs_sommerfeld.pdf \
  --metrics-csv docs/z4c/figures/tde_stellar_morphology_metrics.csv
```

The late CPBC projection log has SHA-256

`24ddd2d0cdba5fe7e975d785e8c4d2a7ccd7ae6b0f1644e2dd1c5c4298105856`.

The causal Sommerfeld and CPBC projection logs have SHA-256 values

`380c58c27faf947ff36d0f994ac18d503d245c1dab7bd2e380f477c29f99af49`
`f04b710b4fcf1bc466b680b8a6d1204b7b140ee5d533f1558548b203e14ae71d`

for Sommerfeld and CPBC, respectively.

8 Conclusion

In the matched no-sponge thin-z TDE experiment, extrapolated Sommerfeld causes a causally timed, lower-boundary-localized instability and becomes nonfinite. The archived zero-rate CPBC removes that failure mode and evolves cleanly to $t/M = 30$. The appropriate claim is therefore:

> **The archived zero-rate CPBC cures the observed Sommerfeld instability in this configuration.**

This is not yet a general CPBC acceptance result or a validation of the newer tangential-principal candidate.